i.mixer

ii.grinders

iii.vacuum cleaners

iv.fans

v.washing machines

vi.centrifugal pumps

vii.egg beaters

viii.Hair drier and

ix.blowers.

8.4.1 Why Single Phase Induction Motor' is not a Self Starting? How to Make Single Phase Induction Motor Self-Starting?

Single phase induction motor has distributed stator winding and squirrel cage rotor. When fed from a single phase supply, its stator winding produces a flux which is only alternating. It is not a synchronously revolving (or) rotating flux as in the case of a two or three phase stator winding fed from a two or three phase supply. Now, an alternating or pulsating flux acting on a stationary squirrel cage rotor cannot produce rotation (only a revolving flux can produce rotation).

To make itself starting, it can be temporarily converted into a two phase induction motor while starting. This can be achieved by introducing an additional starting winding also called as auxiliary winding.

(i) Stator Winding

Stator of a single-phase Induction motor has two windings

1. Main winding (Running winding)
2. Starting winding (auxiliary winding)

These two windings are connected in parallel across single phase supply and are spaced 90° electrical degrees apart, phase difference of 90° electrical degree can be achieved by connecting capacitor in series with the starting winding. Hence the motor behaves like a two phase motor and the stator produces revolving magnetic field which causes rotor to run. Once motor speed, say up to 80% of its normal speed, the starting winding gets disconnected from the circuit by means of a centrifugal Switch and the motor runs only on main winding.

(ii) Rotor

it consists of a laminated cylindrical core having parallel slots on its outer periphery. One copper or aluminum bar is placed in each slot. All these bars are joined at each end by metal ring called end rings. The entire construction resembles like a squirrel cage and hence the name called like that. The rotor is not connected electrically to the supply but has current induced in it by transformer action from the stator.

8.4.2 Classification of Single Phase Motor

a) Split phase induction motor
b) Capacitor start capacitor run motor
c) Universal motor

a) Split Phase Induction Motor

As the starting torque of this type of motor is relatively small and it's starting current is high. These motors are most commonly used for rating up to 0.5 HP where the load could be starting easily

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